Research Report

Expanded Churn Analysis Report

Contents - Executive summary - Visualizations (embedded): Age histogram, Balance histogram (zoomed), Estimated Salary histogram (zoomed), Credit Score histogram - Feature importance charts: Random Forest importances, Permutation importances - SHAP: Not available in this environment; used permutation importance as substitute - Actionable recommendations and target lists

Executive summary (Concise summary repeated from earlier: dataset 10,000 rows; churn 20.37%; model AUC 0.85; key predictors Age, Salary, CreditScore, Balance; high churn in Germany; inactive members likely to churn; older/high-balance customers high value and at risk.)

Visualizations (files saved) - churn_plots/age_by_churn.png — Age distribution by churn - churn_plots/balance_by_churn.png — Balance distribution by churn (zoomed to 99th percentile) - churn_plots/salary_by_churn.png — Estimated Salary distribution by churn (zoomed to 99th percentile) - churn_plots/creditscore_by_churn.png — Credit Score distribution by churn

Feature importance charts - churn_plots/feature_importances.png — Random Forest feature importances (Age highest, then EstimatedSalary, CreditScore, Balance, NumOfProducts) - churn_plots/permutation_importances.png — Permutation importances (robustness check)

SHAP note - The shap library was not available in the execution environment, so I used permutation importance as a robust alternative. If you want SHAP visuals, I can produce them if you allow me to install shap or run in an environment where shap is available.

Target lists and sample - churn_plots/high_precision_targets_germany.csv — 596 customers matching the rule: Geography==Germany & Age>=40 & IsActiveMember==0 - churn_plots/sample_20_rows.csv — random sample of 20 rows for inspection

Next steps I can take - Insert these plots into a single comprehensive PDF with explanations and captions. - If you want SHAP plots specifically, I can attempt to install shap or run the SHAP code elsewhere; alternatively provide an environment that has shap. - Re-run model optimizing for recall, generate lift/ROC curves, and include in the PDF.

Deliverable I will now create a single expanded PDF that embeds the generated charts and updated write-up. Confirm you'd like me to proceed to bundle these images and the analysis into an expanded PDF named 'Churn_Analysis_With_Visuals.pdf'.